

PFTM — Prediction Formation Traceability Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Formation Standard (PFOS)

Operational Layer: Prediction Formation Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Formation Traceability

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ ·
AIG® · ASGA™ · CLIA® ·

SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Prediction Formation Traceability

Canonical Definition

Prediction Formation Traceability Module (PFTM) defines the governance conditions under which a formed prediction remains traceably associated with the predictive components, methodology, Prediction Basis, Predictive Assumptions, applicable conditions, Predictive Claim, Prediction Output, Formation State, provenance, and Formation Time from which it emerged.



FTM establishes the reconstructable lineage of the prediction at the moment of formation.

It transforms:

Governed Prediction Composition + Identified Predictive Claim + Governed Prediction Output + Governed Prediction Formation State

into:

Governed Prediction Formation Trace

The purpose of PFTM is to ensure that a prediction can later answer:

How did this specific prediction come into existence, from what governed components, under what conditions, and in what state?

Its foundational distinction is:

Prediction Record Exists ≠ Prediction Formation Is Traceable

Operational Role

PFTM serves as the fifth and final internal governance space of the Prediction Formation Standard (PFOS) .

It receives:

Governed Prediction Composition

Identified Predictive Claim

Governed Prediction Output

Governed Prediction Formation State

and creates the traceable formation structure linking the formed prediction to its governed origin.

The module asks:

Can this prediction later be reconstructed as it existed at Formation Time, including what formed it, what it meant, what conditions applied, and what state it held?

The governed progression is:

Formation Components Identified → Formation Relationships Linked → Provenance Preserved → Formation Time Anchored → Formation State Captured → Historical Formation Snapshot Preserved → Governed Prediction Formation Trace

The primary output is:

Governed Prediction Formation Trace

This completes the PFOS internal formation sequence and enables the prediction to proceed into:

PPUS — Prediction Probability & Uncertainty Standard.

Internal Module Flow

Prediction Composition → Predictive Claim Identification → Prediction Output Formation → Prediction State Establishment → Prediction Formation Traceability

Current Position: Prediction Formation Traceability

Module Operational Space

PFTM governs:

Prediction Formation trace; Prediction Formation lineage; Formation Time; formation-source association; Prediction Object reference; Prediction Purpose reference; Prediction Horizon reference; Predictive Boundary reference; methodology reference; Prediction Basis reference; Predictive Assumption reference; applicable-condition reference; Prediction Composition reference; Predictive Claim reference; Prediction Output reference; Formation State reference; predictor/source reference; human-contribution reference;

machine-contribution reference; hybrid contribution; transformation lineage; adjustment lineage; selection lineage; formation authority reference where applicable; provenance; version relationships; Formation-Time snapshot; reconstructability; lineage continuity;

missing-link detection; traceability gaps; correction trace; and historical formation integrity.

Module Function

A prediction may be fully formed and still be difficult to reconstruct later.

An organization may know the final output but no longer know:

which model version produced it; which dataset was used; which assumptions were active; whether an analyst changed the model result; what Prediction Horizon applied; which conditions were in force; what the actual claim meant; when the prediction was formed.

PFTM prevents the prediction from becoming detached from its formation context.

It creates the governed trace connecting:

what the prediction became

to:

what caused it to become that prediction.

Prediction Formation Trace

A Prediction Formation Trace is the governed relationship structure connecting a formed prediction to the material components, states, conditions, actions, and references that existed at Formation Time.

The trace is not merely a list of documents.

It represents the lineage of formation.

Traceability ≠ Documentation Volume

An organization may hold thousands of pages of documentation and still be unable to reconstruct a specific prediction.

PFTM therefore distinguishes:

documentation existence

from:

formation traceability.

More Documentation ≠ Greater Traceability by Default

Prediction Identity Reference

The Formation Trace must remain associated with the specific Prediction Identity.

Traceability must not rely solely on:

document title;

model name; dashboard location; date range.

Where multiple predictions exist, each must remain distinguishable.

Prediction Object Reference

The trace should identify the governed Prediction Object from POPS.

This prevents later confusion about what was actually being predicted.

Prediction Purpose Reference

The Prediction Purpose active at Formation Time must remain reconstructable.

A prediction formed for:

exploratory analysis

must not later be represented as though it had been formed for:

automated operational execution.

Current Use $\neq$ Original Prediction Purpose

Prediction Horizon Reference

The Formation Trace must preserve the Prediction Horizon used in the formed prediction.

Later systems must not infer the Horizon from publication date or outcome date.

Predictive Boundary Reference

Where material, the trace preserves the applicable:

geography; population; system; organization; market; asset; scenario; operating environment.

Methodology Reference

The prediction must remain traceably connected to the specific Prediction Methodology used at Formation Time.

This may include:

methodology identifier; methodology version; applicable method state; methodological conditions.

Current Methodology Version $\neq$ Formation-Time Methodology Version

Prediction Basis Reference

The trace preserves which governed Prediction Basis supported the prediction.

This may include references to:

information; datasets; signals; observations; evidence references; source relationships.

PFTM does not re-govern the basis.

It preserves its role in formation.

Formation-Time Basis Snapshot

Later changes to source data must not silently alter the basis associated with the historical prediction.

Current Data $\neq$ Formation-Time Prediction Basis

Predictive Assumption Reference

The Formation Trace preserves the governed assumptions active when the prediction was formed.

This includes the relevant assumption state where material.

Example:

Supplier Continuity Assumption — Conditionally Holding

Later changes to the assumption must not rewrite the earlier state.

Current Assumption State $\neq$ Formation-Time Assumption State

Applicable Conditions Reference

Formation conditions must remain traceable.

These may include:

scenario conditions; operating conditions; regime conditions; threshold conditions; temporal conditions; dependency conditions.

Prediction Composition Reference

PFTM preserves which predictive components were included in the composition and their material roles.

This allows later reconstruction of:

what actually went into the prediction.

Predictive Claim Reference

The exact Predictive Claim identified under PCIM must remain linked to the Formation Trace.

This protects against later reinterpretation.

Prediction Output Reference

The Governed Prediction Output must remain linked to:

claim; composition; formation state; Formation Time.

Formation State Reference

The state assigned under PSEM must remain preserved.

Examples:

Formed

Conditionally Formed

Restricted

The historical prediction must not later appear to have been fully formed if it was originally restricted.

Formation Time

Formation Time represents the governed time at which the prediction became formed in its applicable Formation State.

Formation Time is a core traceability anchor.

It distinguishes:

what was knowable then

from:

what became known later.

Formation Time ≠ Publication Time

A prediction may be formed before it is:

displayed; communicated; approved; published; relied upon.

The trace must preserve this distinction.

Formation Time ≠ Model Execution Time

A model may execute repeatedly.

The Prediction Formation Event may occur only when one output is selected, transformed, approved, or otherwise established as the prediction.

Model Run Time $\neq$ Prediction Formation Time

Formation Time $\neq$ Outcome Time

Outcome information must never be allowed to alter the recorded formation state.

Formation Event Identifier

Where practical, the Formation Trace should assign a distinguishable Formation Event identifier.

This allows:

repeated predictions; versions; revisions; parallel scenarios
to remain separable.

Predictor / Source Reference

The trace may identify who or what materially originated or formed the prediction.

Possible sources include:

individual; team; model; AI system; forecasting platform; hybrid workflow.

Predictor identity does not replace methodology traceability.

Predictor Identity $\neq$ Prediction Methodology

Human Contribution Trace

Where humans materially contribute, PFTM may preserve:

selection; adjustment; override; expert judgment; approval; condition setting.

This allows later distinction between:

model output

and:

final human-influenced prediction.

Machine Contribution Trace

Where machines materially contribute, PFTM may preserve:

model identifier; model version; computational pipeline reference; automated transformation; machine-generated output.

Hybrid Formation Trace

Hybrid predictive processes should preserve the material interaction between human and machine contributions.

Example:

Model A $\rightarrow$ System Calibration $\rightarrow$ Analyst Adjustment $\rightarrow$ Committee Selection $\rightarrow$ Final Prediction

The final output alone is not sufficient to reconstruct this chain.

Transformation Lineage

Any material transformation between predictive components and final output should remain traceable.

Examples:

normalization; conversion; aggregation; calibration; weighting; rounding; category mapping.

Adjustment Lineage

Material adjustments should preserve:

pre-adjustment state; adjustment action; post-adjustment state; source where applicable.

Selection Lineage

Where multiple candidate outputs or claims existed, the trace should preserve which one was selected and, where material, which alternatives were not.

Excluded Candidate Preservation

Not every rejected candidate requires permanent detailed preservation.

However, where later interpretation or audit could depend upon selection context, material excluded alternatives should remain identifiable.

Formation Authority Reference

Where organizational governance requires approval or authority to finalize a prediction, PFTM may preserve the applicable authority reference.

It does not create that authority.

GOA™ and AGA™ retain governance of organizational authority and accountability.

Formation Provenance

Formation Provenance represents the traceable origin structure of the formed prediction.

It may include:

Who / What → Using Which Methodology → From Which Basis → Under Which Assumptions → Through Which Formation Logic → Producing Which Claim / Output → In Which State → At What Time

Provenance ≠ Ownership

Traceability of origin does not by itself determine legal or organizational ownership.

Lineage Continuity

Each material step in formation should remain connected sufficiently to prevent unexplained jumps.

Example:

Model Output: 8.2

↓

Final Prediction: 6.4

without a governed transformation or adjustment reference creates a Traceability Gap.

Missing Formation Link

A Missing Formation Link exists where a material relationship in the formation chain cannot be established.

Examples:

output has no methodology reference; final value cannot be linked to any candidate output; human override is known but undocumented; assumptions used cannot be identified; Formation Time is unclear.

Prediction Formation Traceability Gap

A Prediction Formation Traceability Gap exists where the prediction cannot be sufficiently reconstructed from its formation records.

A gap may be:

Minor

Material

Critical

Unable to Determine

Critical Traceability Gap

A gap may be critical where it prevents establishment of:

what was predicted; when it was formed; which methodology applied; which basis applied; which material assumptions applied; which output was actually formed.

Traceability Gap $\neq$ Prediction Error

A prediction may later be correct despite poor traceability.

The governance problem concerns reconstructability.

Reconstructability

Reconstructability means the prediction's formation can later be understood with sufficient fidelity to establish its material original state.

This does not necessarily require exact computational reproduction.

Reconstructability $\neq$ Exact Replication

A historical AI model may no longer be executable.

The prediction may still be reconstructable if the governed record preserves:

model/version; basis; assumptions; claim; output; state; Formation Time; material transformations.

Exact Replication

Exact replication may be possible in some systems, but PFTM does not require it universally.

The architecture requirement is sufficient reconstructability.

Formation-Time Snapshot

PFTM establishes or references a Formation-Time Snapshot representing the material governed state when the prediction was formed.

This may include:

Prediction Object; Purpose; Horizon; Boundary; methodology version; basis state; assumption set/version; conditions; composition; claim; output; Formation State; provenance; Formation Time.

Snapshot $\neq$ Copy of Every Source Record

The snapshot may use immutable references, hashes, identifiers, version pointers, or equivalent governed mechanisms.

The goal is historical reconstruction, not unnecessary duplication.

Current State vs Formation-Time State

A prediction may later undergo:

new data; revised assumptions; new methodology; new output; revision; supersession.

PFTM preserves the original formation state.

Current Prediction State $\neq$ Formation-Time Prediction State

Historical Reinterpretation Risk

Later knowledge can make an earlier prediction appear:

more obvious; less reasonable; more accurate; less accurate
than it appeared at Formation Time.

PFTM protects against this by preserving the original predictive context.

Later Knowledge $\neq$ Formation-Time Knowledge

This distinction is critical for fair predictive performance assessment.

Formation Trace Version

Material corrections to the trace itself should remain versioned.

Example:

A missing methodology reference is later discovered and added.

The trace may be corrected without rewriting what the original prediction was.

Trace Correction $\neq$ Prediction Revision

Correcting metadata or trace references does not automatically mean the prediction itself changed.

The correction must remain distinguishable from prediction-level revision.

Formation Record Integrity

The trace should resist silent alteration.

Implementations may use:

version control; immutable logs; append-only records; event history; cryptographic verification; signed records.

PFTM does not mandate one specific technology.

Append-Only Principle

Where feasible, material historical formation states should be preserved through append-only or equivalent continuity mechanisms.

New Trace Information $\neq$ Permission to Rewrite Historical Formation State

Formation Trace and PPUS

PPUS receives the formed prediction after PFOS.

Probability and uncertainty governance may add new governed information.

This does not change the original formation trace unless the Predictive Claim or Output is reformulated through a governed process.

Formation Trace and PMRS

PMRS later governs prediction-level revision, supersession, and withdrawal.

PFTM provides the original formation baseline against which later versions can be distinguished.

Formation Trace and POFPS

POFPS later assesses performance against a governed outcome reference.

Without PFTM, performance assessment may suffer from hindsight substitution.

PFTM helps ensure that the prediction compared with the outcome is the prediction that actually existed.

Formation Trace and PRCS

PRCS governs complete Prediction Record & Historical Continuity across the full lifecycle.

PFTM owns formation-stage traceability only.

PRCS preserves the broader lifecycle record.

Formation Traceability $\neq$ Complete Prediction Lifecycle Continuity

Traceability and External Audit

PFTM can support external or internal assessment by making the formation path reconstructable.

However, PFTM itself does not perform an audit.

ADIT® and applicable assessment mechanisms retain those roles.

Traceability and Machine-Readable Governance

A machine-readable Formation Trace may include identifiers for:

Prediction; Formation Event; Claim; Output; Methodology; Basis; Assumption Set; Conditions; State; human/machine contributors; Formation Time; version relationships.

Machine-Readable Formation Graph

Implementations may represent the trace as a graph:

Prediction → formed from → Composition

Composition → uses → Methodology

Composition → uses → Prediction Basis

Composition → depends upon → Assumption Set

Prediction → expresses → Predictive Claim

Prediction → represented by → Prediction Output

Prediction → held state → Formation State

Prediction → formed at → Formation Time

This allows machine systems to traverse formation lineage.

Traceability Governance Sufficiency

Prediction Formation Traceability is sufficiently governed when:

Prediction Identity is preserved; Formation Event is identifiable; Formation Time is established; Prediction Object/Purpose remain linked; Horizon/Boundary remain linked; methodology version remains identifiable; Formation-Time Prediction Basis remains identifiable; Formation-Time Assumption Set remains identifiable; applicable conditions remain identifiable; Composition remains traceable; Predictive Claim remains linked; Prediction Output remains linked; Formation State remains linked; material human/machine contributions remain identifiable; material transformations and adjustments remain traceable; missing links remain explicit; Formation-Time state remains reconstructable; trace corrections remain distinguishable from prediction revisions.

Governance Requirements

PFTM requires that:

1. every formed prediction possess a distinguishable Formation Trace; 2. the trace remain associated with Prediction Identity; 3. Formation Time be identifiable; 4. Formation Time remain distinguishable from publication, model-run, and outcome time; 5. Prediction Object and Purpose remain traceable; 6. Prediction Horizon remain traceable; 7. Predictive Boundary remain traceable where material; 8. applicable Prediction Methodology and version remain identifiable; 9. Formation-Time Prediction Basis remain identifiable; 10. Formation-Time Predictive Assumptions remain identifiable; 11. applicable predictive conditions remain traceable; 12. Prediction Composition remain reconstructable; 13. Identified Predictive Claim remain linked; 14. Governed Prediction Output remain linked; 15. Formation State remain linked; 16. predictor/source remain identifiable where material; 17. material human contributions remain traceable; 18. material machine contributions remain traceable; 19. hybrid contribution paths remain reconstructable; 20. material transformations remain traceable; 21. material adjustments remain traceable; 22. material selection decisions remain traceable; 23. Missing Formation Links remain explicit; 24. Traceability Gaps remain classified where material; 25. reconstructability not require exact replication by default; 26. Formation-Time snapshot remain historically preservable; 27. later information not silently rewrite historical formation state; 28. trace corrections remain distinguishable from prediction revision; 29. historical formation integrity remain protected; 30. formation-stage traceability remain distinguishable from complete lifecycle continuity.

Governance Logic

PFTM applies the following governance logic:

Governed Prediction Composition

Identified Predictive Claim

Governed Prediction Output

Governed Prediction Formation State

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Prediction Identity Associated

↓

Formation Event Identified

↓

Formation Time Established

↓

Prediction Object / Purpose / Horizon / Boundary Linked

↓

Methodology / Version Linked

↓

Formation-Time Prediction Basis Linked

↓

Formation-Time Assumption Set / Conditions Linked

↓

Composition / Claim / Output / State Linked

↓

Human / Machine / Hybrid Contributions Linked where material

↓

Transformations / Adjustments / Selection Linked

↓

Missing Formation Links Assessed

↓

Formation-Time Snapshot Established

↓

Historical Formation Integrity Preserved

↓

Governed Prediction Formation Trace

↓

PPUS — Prediction Probability & Uncertainty Standard

Where sufficient traceability cannot be established:

Prediction Formation Traceability Gap → Clarification / Link Recovery / Qualification / Restriction /
Historical Gap Preservation

Minimum Implementation Framework

A minimum implementation of PFTM must establish:

A. Prediction Identifier

The specific formed prediction.

B. Formation Event Identifier

The applicable Prediction Formation event.

C. Formation Time

The governed time at which the prediction entered its applicable Formation State.

D. Prediction Object & Purpose Reference

The applicable POPS references.

E. Prediction Horizon & Boundary Reference

The applicable temporal and predictive scope.

F. Methodology Reference

Including methodology version where applicable.

G. Prediction Basis Reference

The Formation-Time basis or governed reference set.

H. Predictive Assumption Set Reference

Including Formation-Time assumption state or version.

I. Applicable Conditions Reference

Any material formation conditions.

J. Prediction Composition Reference

The applicable PCOM output.

K. Predictive Claim Reference

The applicable PCIM output.

L. Prediction Output Reference

The applicable POFM output.

M. Formation State Reference

The applicable PSEM output.

N. Contributor / Provenance Record

Where material:

Human / Machine / Hybrid / External / Other

O. Transformation / Adjustment / Selection Trace

Any material formation steps affecting the final output.

P. Traceability Gap State

At minimum:

None / Minor / Material / Critical / Unable to Determine

Q. Formation-Time Snapshot / Historical Reference

The mechanism allowing reconstruction of the prediction as formed.

R. Trace Version

The applicable trace version or equivalent historical identifier.

Governance Outputs

PFTM may produce:

Governed Prediction Formation Trace; Prediction Formation Record; Formation Event Identifier; Formation Time Record; Prediction Object/Purpose linkage; Horizon/Boundary linkage; Methodology/version linkage; Formation-Time Prediction Basis linkage; Formation-Time Assumption linkage; Condition linkage; Prediction Composition linkage; Predictive Claim linkage; Prediction Output linkage; Formation State linkage; predictor/source record; human-contribution trace; machine-contribution trace; hybrid-formation trace; transformation lineage; adjustment lineage; selection lineage; Missing Formation Link Record; Prediction Formation Traceability Gap; Formation-Time Snapshot;

Trace Correction Record; Trace Version; historical formation record.

The primary governed output is:

Governed Prediction Formation Trace

which completes PFOS and supports transition to:

PPUS — Prediction Probability & Uncertainty Standard.

Critical Distinctions

Prediction Record Exists ≠ Prediction Formation Is Traceable

More Documentation ≠ Greater Traceability by Default

Current Use ≠ Original Prediction Purpose

Current Methodology Version ≠ Formation-Time Methodology Version

Current Data ≠ Formation-Time Prediction Basis

Current Assumption State ≠ Formation-Time Assumption State

Formation Time ≠ Publication Time

Model Run Time ≠ Prediction Formation Time

Formation Time ≠ Outcome Time

Predictor Identity ≠ Prediction Methodology

Provenance ≠ Ownership

Traceability Gap ≠ Prediction Error

Reconstructability ≠ Exact Replication

Snapshot ≠ Copy of Every Source Record

Current Prediction State ≠ Formation-Time Prediction State

Later Knowledge ≠ Formation-Time Knowledge

Trace Correction ≠ Prediction Revision

New Trace Information ≠ Permission to Rewrite Historical Formation State

Formation Traceability ≠ Complete Prediction Lifecycle Continuity

Enterprise Application

PFTM provides the historical formation backbone needed for serious Prediction Governance.

Without it, an organization may preserve a final number while losing the predictive context that gave the number meaning.

Months or years later, it may be impossible to determine:

which model version was used; which data supported it; what assumptions were active; whether humans adjusted the output; what Horizon applied;

what conditions governed the prediction; which claim was actually formed; whether the prediction was Formed, Conditional, or Restricted.

PFTM makes the formation event reconstructable before the prediction enters probability, feasibility, communication, monitoring, revision, and performance governance.

Use Case 1 — A Highly Accurate Forecast Cannot Be Reconstructed

A company reviews an AI forecast issued twelve months earlier.

The historical dashboard shows:

Predicted demand: 84,000 units

Actual demand was:

83,600 units.

The prediction appears extremely accurate.

However, the company cannot establish:

which model version produced 84,000; whether analysts adjusted the output; which customer dataset was available at Formation Time; which assumptions were active; whether the prediction referred to national or regional demand.

The critical governance question is:

Can observed accuracy demonstrate predictive performance when the organization cannot reconstruct how the prediction was actually formed?

PFTM requires the prediction to remain linked to its Formation-Time methodology, basis, assumptions, claim, output, scope, contributors, and state.

The later performance assessment can therefore evaluate the prediction that actually existed rather than reconstructing a convenient version after the outcome became known.

Observed Accuracy $\neq$ Reconstructable Predictive Performance

Use Case 2 — A Model Appears to Have Predicted a Failure Before It Happened

An industrial company investigates a major equipment failure.

Its AI platform contains a historical prediction record showing:

Failure risk: High

dated three days before the incident.

The company initially concludes that the system successfully predicted the failure.

Further review shows:

the model produced an anomaly output three days before failure; an analyst converted that anomaly into a “High Failure Risk” label only after the incident; the system retained the original model-run date as the displayed prediction date.

The critical governance question is:

Did the governed failure prediction actually exist before the failure, or was later interpretation attached retrospectively to an earlier model output?

PFTM separates:

Model Run Time

from:

Prediction Formation Time

and preserves the historical relationship between the original output, later human interpretation, and final Predictive Claim.

A later-created prediction cannot acquire an earlier Formation Time merely because it was derived from earlier data.

Later Interpretation $\neq$ Earlier Prediction Formation

Canonical Boundary

PFTM governs the traceable association, provenance, lineage, Formation Time, Formation Event, contributor relationships, transformation history, version linkage, Formation-Time snapshot, reconstructability, Traceability Gap identification, correction trace, and historical integrity of a prediction at formation. It does not govern Prediction Composition itself, Predictive Claim Identification itself, Prediction Output Formation itself, Formation State determination itself, probability and uncertainty representation, Prediction Feasibility, Collective Prediction governance, communication and reliance, prediction-level monitoring and revision, outcome determination, predictive performance assessment, or complete lifecycle record continuity.

PFTM establishes:

how the formed prediction remains connected to the governed state from which it originated.

It does not establish:

whether the prediction will be correct or how its complete lifecycle history will later evolve.

Canonical Closing Statement

A prediction whose origin cannot be reconstructed becomes vulnerable to hindsight, reinterpretation, and false attribution.

PFTM preserves the formation path so that methodology, basis, assumptions, claim, output, state, contributors, and time remain attached to the prediction that actually existed.

A prediction should be judged from the state in which it was formed — not from the story reconstructed after the future became known.

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contact@originopenfoundation.org